Indian Institute of Technology Palakkad

Algorithms

Assignment: Lists, Stacks & Queues

1. Explain how to implement two stacks in one array of size n in such a way that neither stack overflows unless the total number of elements in both stacks together is n. The PUSH and POP operations should run in constant time.

2. Show how to implement a queue using two stacks. Analyze the running time of the queue operations.

3. Show how to implement a stack using two queues. Analyze the running time of the stack operations.

4. While a stack allows insertion and deletion of elements at only one end, and a queue allows insertion at one end and deletion at the other end, a deque (double-ended queue) allows insertion and deletion at both ends. Write four constant-time procedures to insert elements into and delete elements from both ends of a deque implemented by an array.

5. Give a linear-time nonrecursive procedure that reverses a singly linked list of elements. The procedure should use no more than constant storage beyond that needed for the list itself.